

Synaptic Depression Modeling

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Abstract

We describe a real-time adaptive Bernoulli state-space model (BSSM) for a network of interacting neurons with short-term synaptic depression (STD) and a lag- M synaptic memory kernel. The STD mechanism is represented as a latent dynamical resource variable that is updated at each discrete time step, coupled to observed binary spike indicators through a Bernoulli likelihood. For context, we also briefly review the Tsodyks–Markram (TM) resource model, which provides the continuous-time biological foundation for the discrete STD dynamics used in the BSSM.

1 Short-Term Synaptic Depression (STD) in Cortical Synapses

Short-term synaptic depression (STD) is a fast, reversible decrease in synaptic strength that occurs when a presynaptic cortical neuron fires repeatedly over short time windows (typically from milliseconds to seconds).

Activity-decaying phenomenon. When presynaptic firing is high, the postsynaptic response (EPSCs or IPSCs) becomes progressively smaller for successive spikes, producing a decaying envelope of synaptic efficacy during bursts. After presynaptic activity pauses, synaptic strength gradually recovers to baseline levels.

Main biological mechanism. STD is commonly modeled as depletion of synaptic “resources” required for effective neurotransmitter release. At each presynaptic spike, release depends on (i) the availability of readily releasable vesicles (or release-ready sites) and (ii) an effective probability of release (often related to intracellular calcium dynamics). Sustained activity depletes resources and reduces future release, while periods of silence allow recovery.

Relevant timescales. Recovery is typically on intermediate timescales (~ 100 ms to a few seconds), depending on synapse type and experimental protocol.

Functional consequences. STD acts as a rate-dependent synaptic filter: it attenuates sustained high-rate input, emphasizes transient changes in firing, and contributes to network stabilization by preventing runaway excitation.

2 Computational Models with STD

We describe two STD frameworks. The TM model (Section 2.1) is included for conceptual context only. The BSSM (Section 2.2) is the target implementation model.

The following notation is shared across both models: $U \in (0, 1]$ is the utilization parameter (fraction of available resources used per spike), and τ_{rec} is the STD recovery time constant.

2.1 Tsodyks–Markram (TM) Resource Model: Conceptual Background

The *Tsodyks–Markram (TM)* model [1] is a widely used continuous-time resource model for *short-term plasticity (STP)*. In the STD-only version, the synapse has a single *available-resource* variable $x(t) \in [0, 1]$. Between presynaptic spikes, resources recover exponentially:

$$\frac{dx}{dt} = \frac{1 - x}{\tau_{\text{rec}}}. \quad (1)$$

At a presynaptic spike time $t = t_k$, a fraction U of available resources is consumed:

$$x(t_k^+) = x(t_k^-) (1 - U), \quad (2)$$

and the released neurotransmitter amplitude at that spike is

$$r(t_k) = U x(t_k^-). \quad (3)$$

The TM model provides the biological motivation for the discrete STD resource update used in the BSSM below (Eq. 5), which is the exact discretization of Eq. (1) combined with Eq. (2).

2.2 Real-Time Adaptive Bernoulli State-Space Network Model (BSSM) with STD and Lag- M Kernel

The *Bernoulli state-space model (BSSM)* represents a network of N neurons using (i) latent dynamical states encoding STD resources and (ii) a Bernoulli observation model for binary spike indicators conditioned on those states. This is the primary model of interest. At physiological firing rates (5–20 Hz) and bin widths ($\Delta t = 1$ –2 ms), the probability of two spikes per bin is of order $p^2/2 \sim 10^{-5}$, making Bernoulli statistically indistinguishable from Poisson while simplifying both simulation and inference.

Discrete time and notation. Let $t \in \{0, 1, \dots, T-1\}$ index time bins of width Δt . For each neuron j , the primary observable is the binary *spike indicator*

$$s_{j,t} = \begin{cases} 1 & \text{neuron } j \text{ fired a spike in bin } t, \\ 0 & \text{neuron } j \text{ was silent in bin } t, \end{cases} \quad (4)$$

which records only *whether* a spike occurred. The downstream efficacy of that spike is determined separately by the latent STD resource variable $x_{j,t} \in [0, 1]$, which encodes the fraction of neurotransmitter available for release at the start of bin t (see Eq. 5). A spike that occurs when resources are depleted ($x_{j,t} \approx 0$) contributes little to postsynaptic drive, whereas a spike after full recovery ($x_{j,t} \approx 1$) contributes maximally.

STD latent state dynamics. Each presynaptic neuron j has its own latent resource variable $x_{j,t}$, tracking the STD state of that neuron’s outgoing synapses independently of all other neurons. The full network state is the collection $\{x_{j,t}\}_{j=1}^N$, with each variable evolving as the exact discretization of the TM continuous-time recovery (Eq. 1) combined with spike-triggered consumption (Eq. 2):

$$x_{j,t+1} = \underbrace{\left(1 - (1 - x_{j,t}) e^{-\Delta t / \tau_{\text{rec}}}\right)}_{\text{exponential recovery}} \cdot \underbrace{(1 - U s_{j,t})}_{\text{spike-triggered depletion}}, \quad j = 1, \dots, N. \quad (5)$$

When $s_{j,t} = 0$, resources recover exponentially toward 1; when $s_{j,t} = 1$, a fraction U is consumed. Because the update depends only on neuron j ’s own spike indicator, resource variables of different neurons are conditionally independent given their respective spike histories, allowing all N updates to proceed in parallel.

STD-modulated release. The effective synaptic release from neuron j in bin t is:

$$u_{j,t} = U x_{j,t} s_{j,t}. \quad (6)$$

This quantity is zero in the absence of a spike, small when resources are depleted by recent activity, and large (up to U) after full recovery.

Lag- M synaptic kernel. Synaptic currents have temporal extent beyond a single bin. We convolve the STD-modulated release with a causal lag- M kernel $\kappa = (\kappa_1, \dots, \kappa_M)^\top$, $\kappa_\ell \geq 0$, encoding how a release event at lag ℓ bins contributes to postsynaptic drive at time t . The effective STD-filtered synaptic input from neuron j at time t is:

$$h_{j,t} = \sum_{\ell=1}^M \kappa_\ell u_{j,t-\ell}. \quad (7)$$

The kernel length must satisfy $M\Delta t \gtrsim \tau_{\text{rec}}$ to capture the full STD recovery window. Practical kernel choices include exponential decay, raised cosine basis functions, or a free learnable filter (see Section 3.2 for the generator choice).

Bernoulli observation model. For each postsynaptic neuron i , the conditional spiking probability is:

$$p_{i,t} = \sigma \left(b_i + \sum_{\substack{j=1 \\ j \neq i}}^N W_{ij} h_{j,t} \right), \quad (8)$$

where $\sigma(z) = 1/(1 + e^{-z})$ is the logistic sigmoid, b_i is a baseline log-odds for neuron i , and W_{ij} is the synaptic weight from neuron j to neuron i (positive for excitatory, negative for inhibitory connections). Self-connections are excluded by the constraint $j \neq i$. The spike indicator is then drawn as:

$$s_{i,t} \sim \text{Bernoulli}(p_{i,t}). \quad (9)$$

For $t - \ell < 0$, pre-history values $u_{j,t-\ell}$ are set to zero, corresponding to a fully recovered synapse with no prior spiking.

Summary of model parameters. The full parameter set of the BSSM is $\theta = \{b_i, W_{ij}, \kappa_\ell, U, \tau_{\text{rec}}\}$, where b_i and W_{ij} are neuron- and synapse-specific, while U and τ_{rec} may be shared across all synapses or made neuron-specific depending on modeling assumptions. In the generator (Section 3), all parameters are fixed to known ground-truth values; in inference, they constitute the quantities to be estimated.

Real-time and adaptive aspects. The term *real-time* refers to the fact that both the latent STD state $x_{j,t}$ and parameter estimates can be updated sequentially as new spike indicators $s_{i,t}$ arrive, without reprocessing the full data history. The term *adaptive* refers to the property that $x_{j,t}$ automatically tracks recent presynaptic spiking through Eq. (5): resources deplete during bursts and recover during silence, so the effective synaptic gain $u_{j,t}$ adapts online to the presynaptic firing history. Together, these properties make the BSSM suitable for closed-loop neural decoding or real-time inference scenarios.

3 Data Generator: Mammalian Cortical Network with STD

To validate the BSSM and test inference procedures, we generate synthetic spike trains from a network that mimics key properties of mammalian cortical dynamics. The generator uses the same BSSM structure as the inference model (Section 2.2), but with all parameters fixed to known ground-truth values, enabling direct comparison between true and estimated quantities.

3.1 Network Architecture

We consider a network of $N \in [50, 500]$ neurons partitioned following Dale’s law into $N_E = 0.8 N$ excitatory and $N_I = 0.2 N$ inhibitory neurons. The code constructs the initial synaptic matrix $W \in \mathbb{R}^{N \times N}$ with a spatial Dale generator rather than an iid Bernoulli mask. Neurons are placed on a two-dimensional grid over $[0, L] \times [0, H]$ with minimum spacing $d_{\min}$, and their order is sorted by position. For each presynaptic neuron j , an out-degree k_j is drawn from the configured range and k_j distinct postsynaptic targets are sampled without replacement with probability proportional to $D_{ij}^{-\delta}$, excluding $i = j$. This gives nearby neurons higher connection probability while keeping the diagonal zero.

The nonzero weights obey Dale’s law column-by-column:

$$W_{ij}^{(0)} = \begin{cases} +a_{ij} & \text{if neuron } j \text{ is excitatory,} \\ -\frac{N_E}{N_I} a_{ij} & \text{if neuron } j \text{ is inhibitory,} \end{cases} \quad a_{ij} \sim \text{Uniform}(1 - v, 1 + v), \quad (10)$$

for sampled edges only, with $W_{ii}^{(0)} = 0$. Finally, the whole matrix is rescaled to the requested spectral radius R ,

$$W = W^{(0)} \frac{R}{\rho(W^{(0)})}, \quad (11)$$

which preserves sparsity, signs, Dale’s law, and the zero diagonal.

3.2 Synaptic Kernel

We use an *exponential decay kernel* as the ground-truth synaptic filter:

$$\kappa_\ell = (1 - \alpha) \alpha^{\ell-1}, \quad \ell = 1, \dots, M, \quad (12)$$

where $\alpha = e^{-\Delta t / \tau_s}$ and τ_s is the synaptic time constant. This kernel corresponds to a standard single-compartment synaptic current model and is the exact discrete-time analog of the continuous-time shot-noise synaptic model. It is normalized so that $\sum_{\ell=1}^M \kappa_\ell \approx 1$ for $M \Delta t \gg \tau_s$.

A key computational advantage of this choice is that the lag- M convolution in Eq. (7) collapses to a single recursive update:

$$h_{j,t} = \alpha h_{j,t-1} + (1 - \alpha) u_{j,t-1}, \quad (13)$$

which avoids explicitly storing M past values of $u_{j,t}$ and scales efficiently to large networks. The decay rate of κ_ℓ is set by the fast synaptic time constant τ_s , whereas the slower STD time constant τ_{rec} enters through the latent state update and the choice of history length M .

3.3 Ground-Truth Parameter Values

Table 1 lists the ground-truth parameter values used in the generator, motivated by mammalian cortical physiology.

3.4 Generation Procedure

Given the fixed parameters, synthetic spike indicators are generated sequentially for $t = 0, 1, \dots, T - 1$ as follows.

3.5 Network Stability Check

With recurrent excitatory connections, the network can exhibit runaway activity if synaptic weights are too large. Before generating long recordings, we recommend a short pilot run (~ 1 s) and verification that mean firing rates across neurons remain in the physiological range of 5–20 Hz. If rates exceed this range, one should reduce σ_W or increase U : stronger STD provides a natural activity-dependent stabilizing mechanism by reducing synaptic efficacy during high-rate episodes.

Parameter	Symbol	Value	Rationale
Time bin width	Δt	1–2 ms	Resolves individual spikes; ensures $p \ll 1$
Synaptic time constant	τ_s	3–5 ms	AMPA-like fast excitation
STD recovery time	τ_{rec}	200–500 ms	Typical cortical STD
Utilization parameter	U	0.3–0.5	Moderate depression
Weight scale	v, R	$v = 0.4, R = 0.90$	Uniform weights scaled by spectral radius
Baseline log-odds	b_i	tuned per neuron	Yields 5–20 Hz firing
Kernel length	M	$\geq \tau_{\text{rec}}/\Delta t$	Covers STD recovery window

Table 1: Ground-truth generator parameters. Ranges reflect variability across synapse types and network sizes. The kernel length M must satisfy $M\Delta t \gtrsim \tau_{\text{rec}}$ to ensure the history window captures the full STD recovery dynamics. The baseline b_i is expressed as a log-odds consistent with the Bernoulli observation model (Eq. 8).

Algorithm 1 Synthetic network spike train generator

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1: Initialize:  $x_{j,0} \leftarrow 1$  for all  $j$  (fully recovered);  $h_{j,0} \leftarrow 0$  for all  $j$ ; draw  $W, \{b_i\}$  as in Eqs. (10)–(11).
2: for  $t = 0, 1, \dots, T - 1$  do
3:   for each neuron  $j = 1, \dots, N$  do
4:     Compute STD-modulated release:  $u_{j,t} \leftarrow U x_{j,t} s_{j,t}$  (Eq. 6)
5:     Update recursive synaptic filter:  $h_{j,t} \leftarrow \alpha h_{j,t-1} + (1 - \alpha) u_{j,t-1}$  (Eq. 13)
6:   end for
7:   for each neuron  $i = 1, \dots, N$  do
8:     Compute spiking probability:  $p_{i,t} \leftarrow \sigma(b_i + \sum_{j \neq i} W_{ij} h_{j,t})$  (Eq. 8)
9:     Draw spike indicator:  $s_{i,t} \sim \text{Bernoulli}(p_{i,t})$  (Eq. 9)
10:  end for
11:  for each neuron  $j = 1, \dots, N$  do
12:    Update STD resource:  $x_{j,t+1} \leftarrow (1 - (1 - x_{j,t})e^{-\Delta t/\tau_{\text{rec}}}) \cdot (1 - U s_{j,t})$  (Eq. 5)
13:  end for
14: end for
15: Output: spike indicators  $\{s_{j,t}\}$ , ground-truth STD states  $\{x_{j,t}\}$ , weight matrix  $W$ .

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References

- [1] M. V. Tsodyks and H. Markram. The neural code between neocortical pyramidal neurons depends on neurotransmitter release probability. *Proceedings of the National Academy of Sciences*, 94(2):719–723, 1997. doi:10.1073/pnas.94.2.719.